



# Benchmarking and Performance Analysis of LLM Inferencing Systems

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# Contents





1 Radix Tree Workload Modes





# Full Sharing (All share 3L)

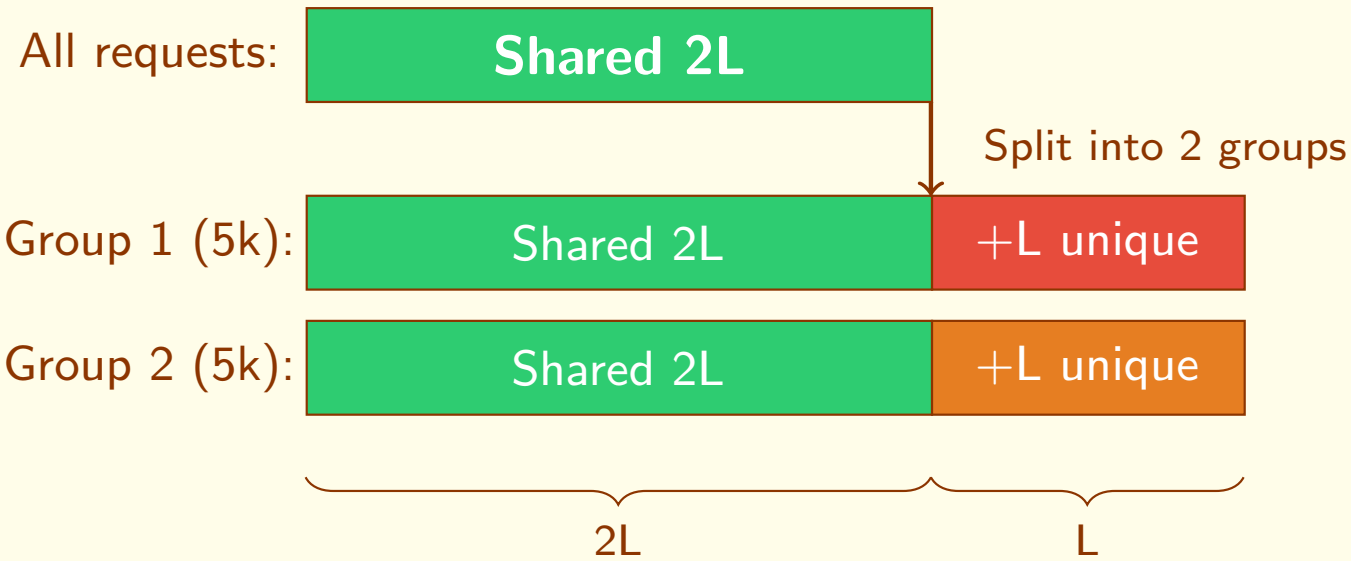
All requests: **Shared 3L**

3L tokens





## All share 2L, then 2xL

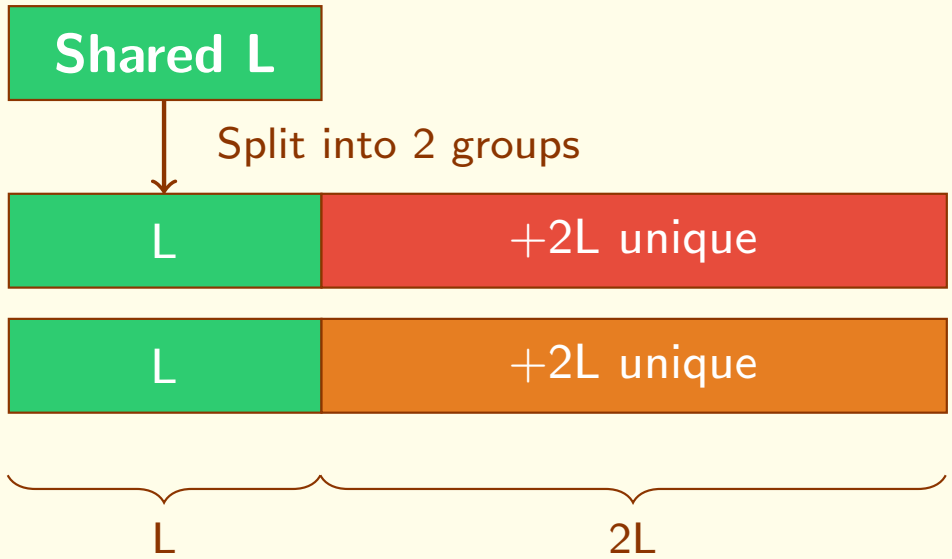






# All share L, then 2x2L

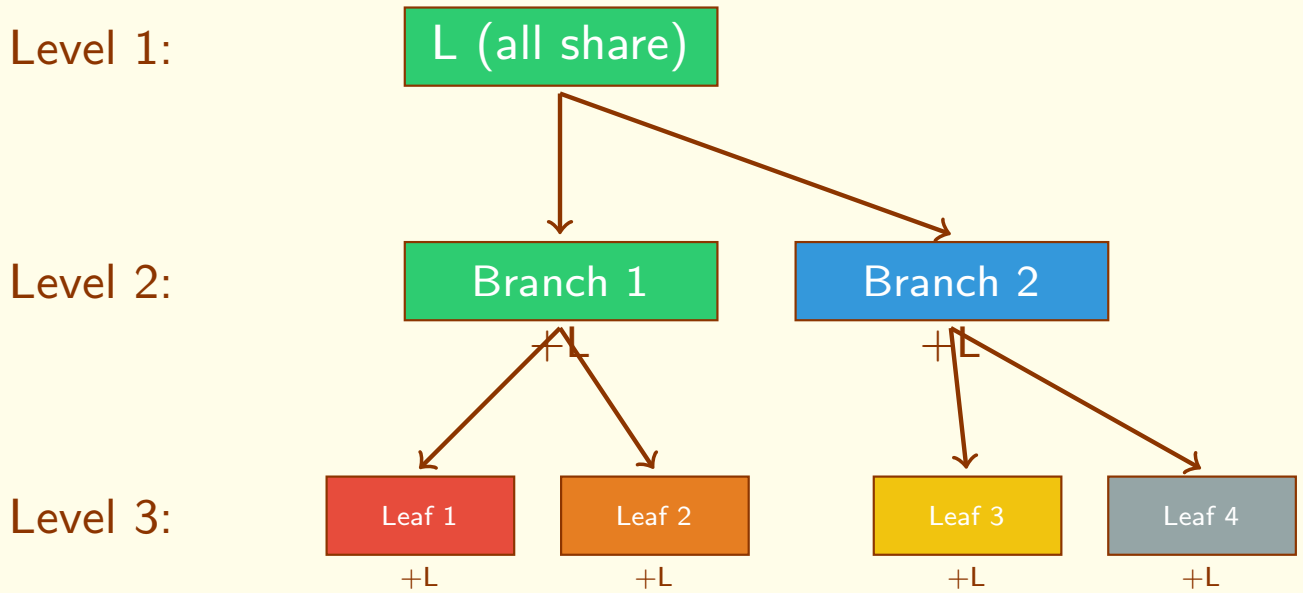
All requests:







# Three-Level Tree (L → 2L → 3L, 4 leaves)







## 2 separate groups with 3L each

Group 1: **Context A (3L tokens)**

Group 2: **Context B (3L tokens)**

3L tokens (completely different)

# Results across varying L

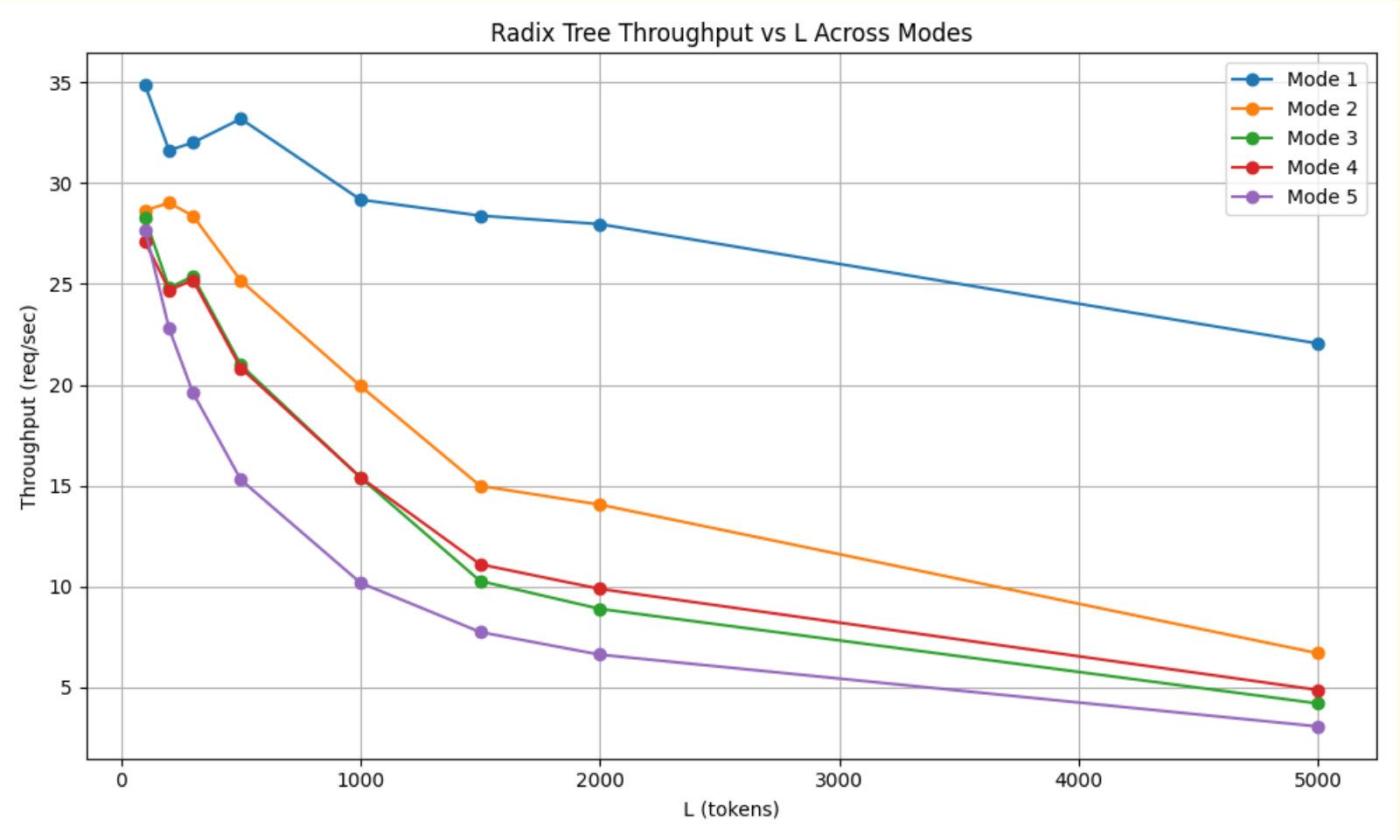
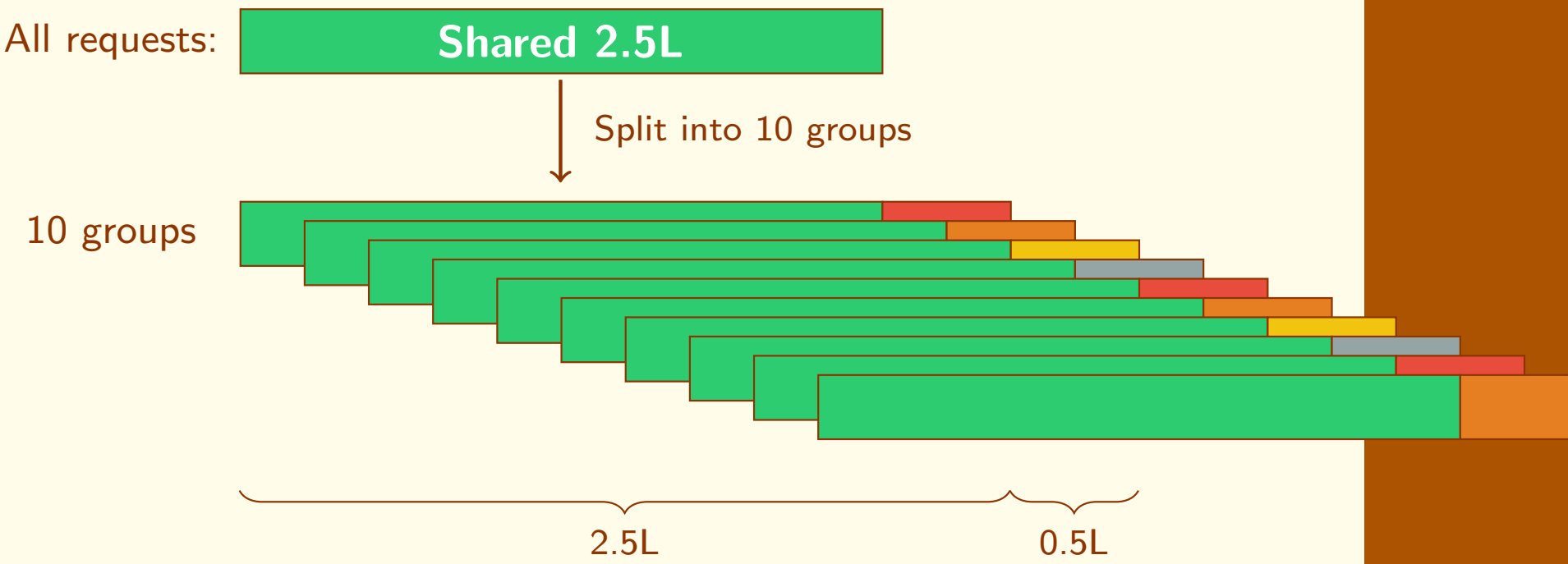



Figure: Results across varying L

# Mode 6



## 2.5L shared + 10x0.5L



# Mode 7

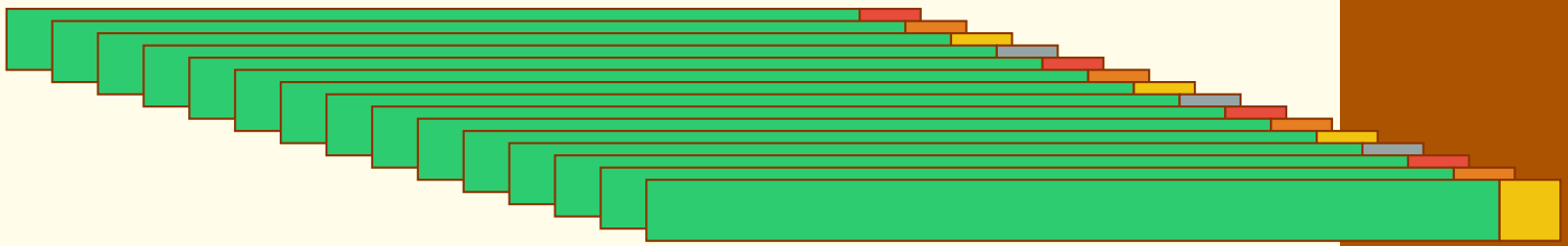




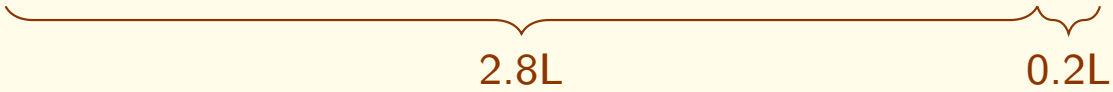
## 2.8L shared + 50x0.2L

All requests: **Shared 2.8L**

50 groups



...(More groups)



# Mode 8

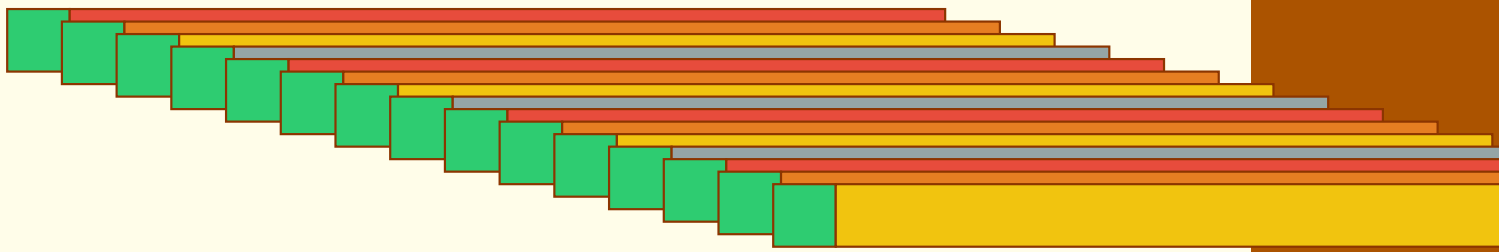




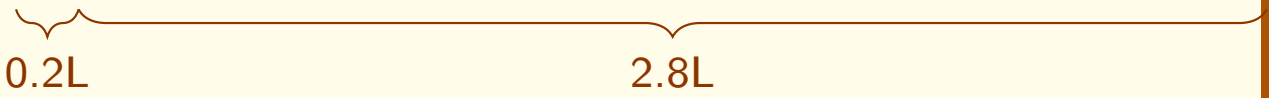
## 0.2L shared + 50x2.8L

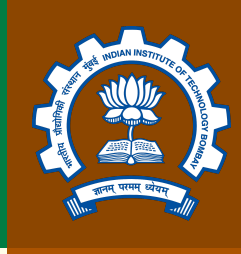
All 10k requests: 0.2L

50 groups



...(More groups)







# Thank You